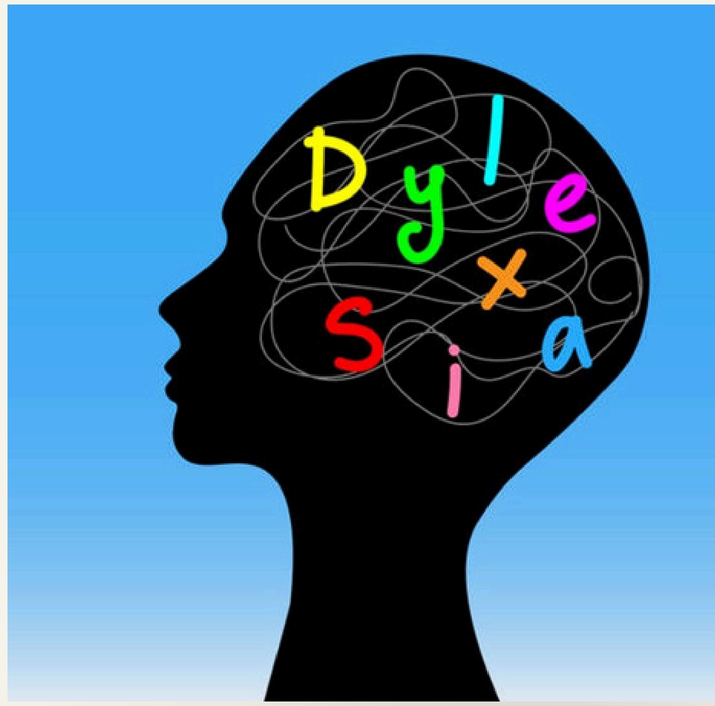
23PCCE501L

Artificial Intelligence and Machine Learning Laboratory

DYSLEXIA DETECTION SYSTEM USING CNN MODEL



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Overview

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Introduction

- Dyslexia is a common neurodevelopmental learning disorder affecting reading, writing, and phonological processing.
- Early detection is critical because timely intervention significantly improves learning outcomes and confidence.
- Many children remain undiagnosed due to limited specialists, time-consuming assessments, and resource-heavy screening methods.
- Advances in Artificial Intelligence, particularly Convolutional Neural Networks (CNNs), provide a promising solution for faster and more accurate dyslexia detection.
- CNN-based tools can analyze visual reading behaviors or handwriting patterns to identify early signs of dyslexia.
- These AI systems can support teachers, parents, and clinicians by offering accessible, scalable, and efficient screening options.

Literature review

- Research evolved from subjective behavioral assessments to computational methods, emphasizing handwriting as a practical and accessible diagnostic source.
- Classical ML on Handwriting Features
- Early models (SVM, k-NN, decision trees) analyzed manually engineered handwriting traits—spacing, letter shape, stroke consistency—but were limited by feature-engineering complexity.
- CNN-Based Handwriting Recognition
- Deep learning, especially CNNs, became the leading approach by automatically learning visual patterns such as irregular strokes, distorted characters, and inconsistent spacing, improving detection accuracy.
- Need for Lightweight CNN Models
- Existing deep-learning methods often require large datasets; this gap motivates developing efficient, small-scale CNN-ML models suitable for limited handwriting datasets—like the approach proposed in this study.

Objective

To detect dyslexia from handwriting images using a lightweight CNN-based model for early, accessible, and accurate screening.

- Collect and preprocess handwriting image datasets from dyslexic and non-dyslexic individuals.
- Design a CNN architecture that extracts key handwriting features such as spacing, stroke irregularities, and letter shape distortions.
- Train and evaluate the CNN classifier to differentiate between dyslexic and non-dyslexic handwriting samples.
- Compare CNN performance with traditional machine learning classifiers (SVM, k-NN, Random Forest).
- Build a lightweight model capable of working effectively with limited datasets.
- Achieve high accuracy, precision, recall, and F1-score to ensure reliable dyslexia screening.
- Develop an accessible, low-cost tool that can support early dyslexia identification in schools and educational settings.

Methodology

Dataset Creation

– 50 handwriting samples collected per class
– Manual verification and labeling

Pre-processing Steps

- Grayscale conversion to simplify pixel intensity information.
- Noise removal using smoothing filters to enhance handwriting clarity.
- Thresholding/Binarization to separate foreground strokes from the background.
- Resizing to 128×128 pixels to maintain uniform input dimensions.
- Normalization (pixel values scaled to 0–1) for stable model training.
- Data augmentation including rotation, zoom, and shifting to increase variability.

Dataset Splitting

The dataset is divided into Training (70%), Validation (15%), and Testing (15%) to ensure unbiased evaluation.

Methodology

Feature Extraction

- Features are extracted directly from images using a Convolutional Neural Network (CNN).
- CNN layers learn high-level handwriting patterns such as curves, spacing, and stroke consistency.

Model Training

- A multi-layer CNN with Conv → ReLU → MaxPooling blocks was used.
- Binary Cross-Entropy loss guides training:
  $L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$
- Adam optimizer updates weights using gradient descent:
  $W = W - \alpha(\partial L/\partial W)$
- Early Stopping avoids overfitting by monitoring validation loss.

Performance Evaluation

- Model performance assessed using:

1. Accuracy, Precision, Recall, F1-Score
2. Confusion Matrix for class-wise error analysis

- Best model exported and integrated into the Streamlit web application for real-time predictions.

Result

I. Handwriting Analysis (CNN Model) 📝

1. **Input**: The user uploads a handwriting image file (e.g., JPG, PNG) via the Streamlit interface.
2. **Preprocessing**: The image is resized to 128 x 128 pixels and normalized (pixel values scaled from 0 to 1).
3. **Prediction**: The pre-processed image is passed to the loaded CNN model (model_cnn.h5).
4. **Score Output:** The model outputs a single Model Score P{Dyslexic} between 0 and 1 (e.g., 0.7845).
5. **Classification**: The score is compared against the 0.5 threshold:
   - If Score >= 0.5: System predicts "Likely Non-dyslexic handwriting detected."
   - If Score < 0.5: System predicts "Likely Dyslexic handwriting detected."

II. Text Analysis (Linguistic Features) ✍️

1. **Input**: The user types or pastes a paragraph of text into the text area.
2. **Spelling Correction:** The input text is processed using TextBlob to generate a corrected version.
3. **Spelling Accuracy** Calculation: The system compares the words in the original input to the corrected text to count words that were initially correct.
   - Result: Spelling Accuracy (%) is displayed. (A lower score suggests more misspellings.)
4. **Grammar Score Calculation**: The score is approximated based on the number of corrections required.
   - Result: Grammar Score (%) is displayed. (A lower score suggests more frequent mechanical errors.)
5. **Correction Display**: The fully corrected text is displayed to the user for reference.

III. Dyslexia Trait Flags and Suggestions 💡

1. **Trait Evaluation**: The system analyzes the results from both handwriting (if analyzed) and text input to trigger simple Dyslexia Trait Flags (e.g., frequent misspellings, short word usage, handwriting inconsistencies).
2. **Trait Output:** A list of Detected Dyslexia Traits is displayed.
3. **Suggestions**: Corresponding Suggestions/Next Steps (e.g., "Handwriting practice," "Use spelling apps") are generated and displayed to the user based on the detected traits.

Feature Extracted	**Method / Clinical Relevance**
Spelling Accuracy	Computed using TextBlob.correct(). A low percentage indicates a high rate of substitution and omission errors, which are common markers of phonological awareness deficits.
Grammar Score	Derived from the percentage of words requiring correction. A lower score suggests more frequent syntactical or mechanical errors.
Dyslexia Trait Flags	Uses basic heuristics (e.g., frequent misspellings, short word usage) to flag preliminary concerns for immediate user feedback.

Result

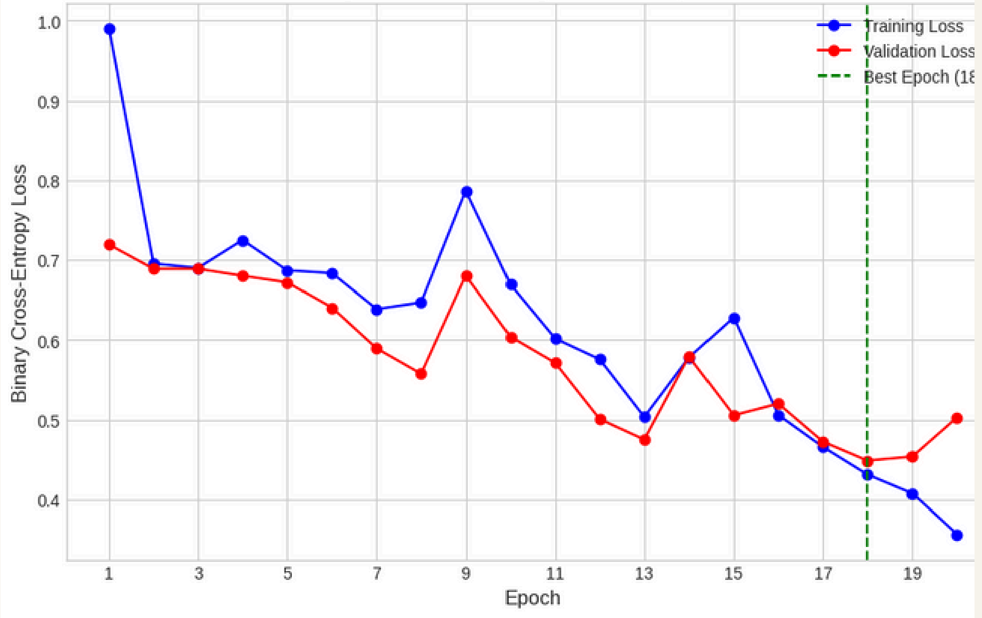


Fig. 4. Model Training Loss over Epochs

This graph tracks the error (Loss) the model makes, aiming to minimize this value during training.

- **Training Loss (Blue Line):** This is the error rate on the data the model actively learns from.
  - Trend: Shows a continuous, steep downward trend (approx 0.99 to approx 0.36), confirming the Adam optimizer successfully learned the patterns and reduced error on the training set.
- **Validation Loss (Red Line):** This is the error rate on the unseen data, making it the most critical metric for generalization.
  - **Trend**: Decreases rapidly until Epoch 18 (Loss approx 0.4487), marking the best point of generalization.
  - **Conclusion (Best Epoch)**: The green dashed line marks Epoch 18. This is the model's peak performance, as it achieved the lowest error on the unseen validation data. The slight increase in loss afterward (Epochs 19-20) suggests that the training process should stop, likely via the Early Stopping mechanism, to retain the optimal weights from Epoch 18.

Metric	Value	Key Insight
Optimal Epoch	**18**	Best performance before overfitting/saturation.
Validation Accuracy	**8.000%**	High correctness on **unseen** handwriting samples.
Validation Loss	**4.487**	Low error rate at the optimal point.

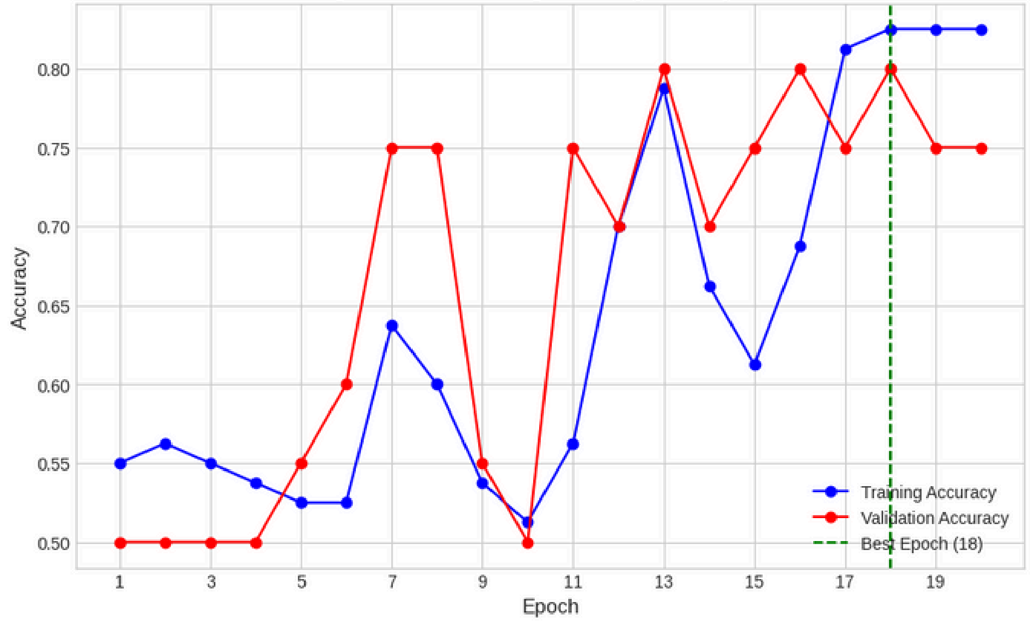
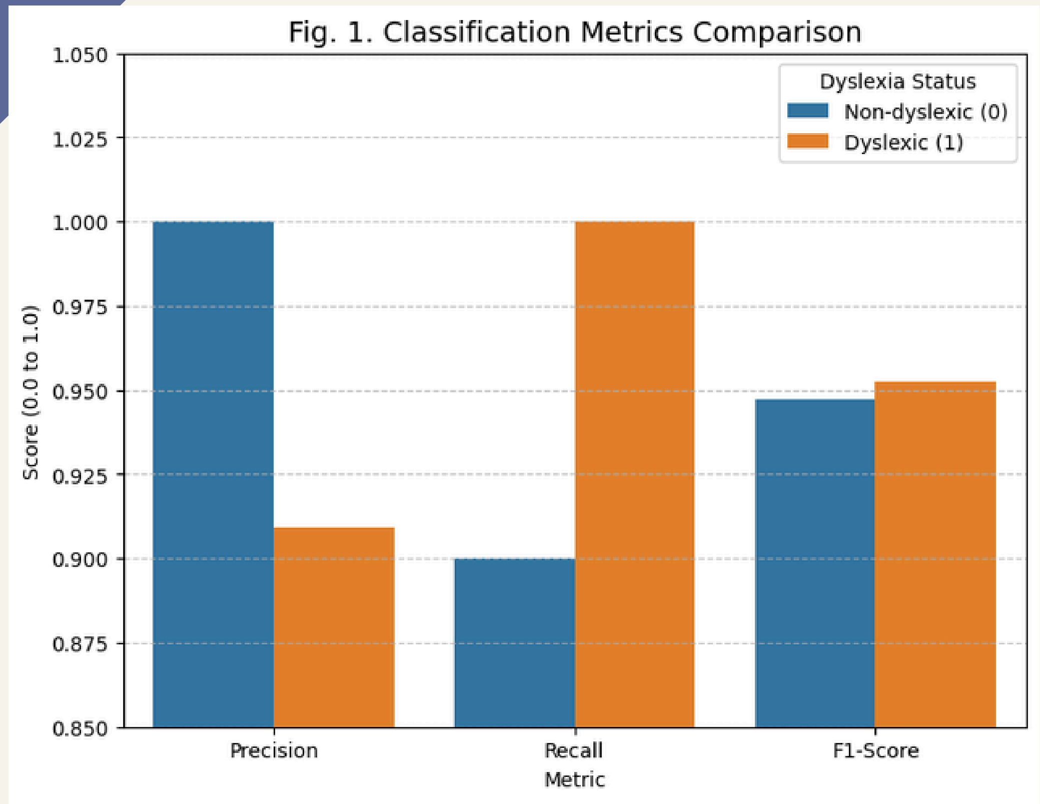


Fig. 5. Model Training Accuracy over Epochs

This graph tracks the correctness (Accuracy) of the model's predictions.

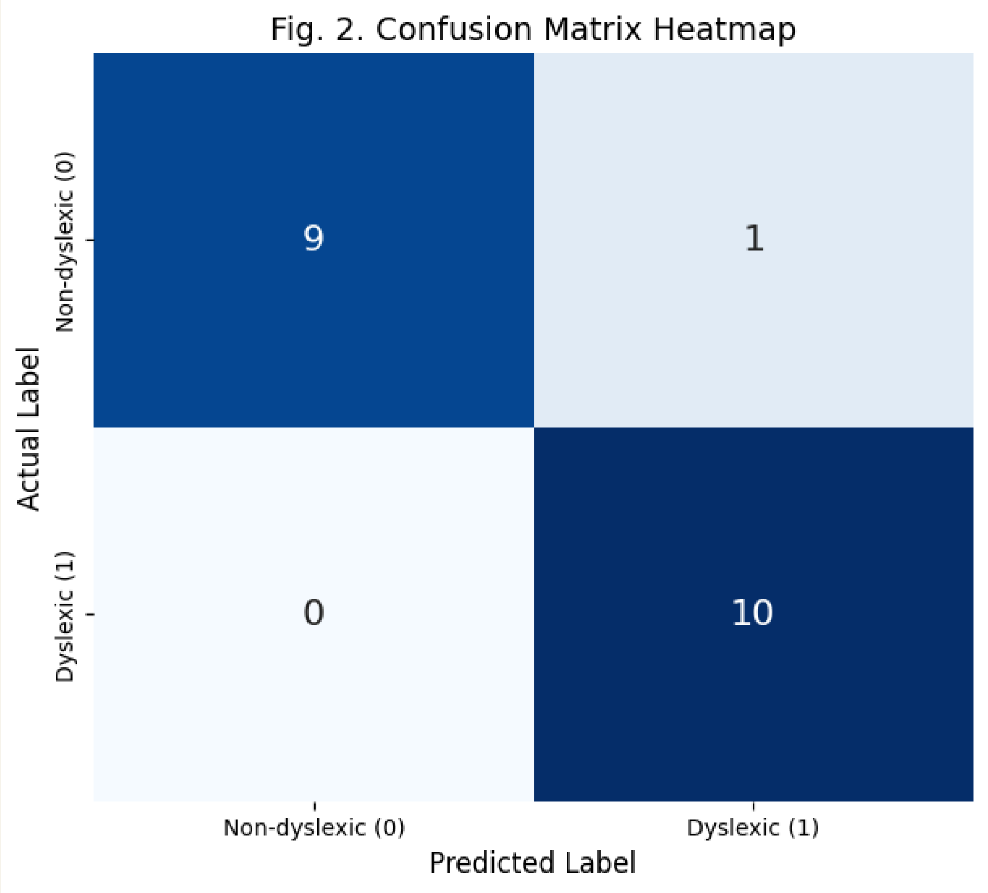
- **Training Accuracy (Blue Line):** Shows a sharp increase (approx 55% to 82.50%), confirming the model rapidly learned the training data.
- **Validation Accuracy (Red Line):** Closely tracks the training curve, peaking at 80.00% around Epoch 18.
- **Overfitting Check:** The small 2.5% gap between training (82.50%) and validation (80.00%) confirms good generalization. This shows the data augmentation and dropout layers successfully prevented severe overfitting.

Result



Explanation of Classification Metrics Comparison(Fig. 1)

- **Safety Priority (Perfect Recall):** The Recall for the Dyslexic (1) class is 1.00 (100%). This is the most critical result, confirming zero False Negatives (FN=0); every true dyslexic case was correctly identified. The model highly prioritizes clinical sensitivity.
- **Accuracy of Negatives:** The Precision for the Non-dyslexic (0) class is also 1.00 (100%), meaning every "Non-dyslexic" prediction was correct.
- **Single Error Trade-off:** The slightly lower Precision for Dyslexic (1) (approx 0.91) and the slightly lower Recall for Non-dyslexic (0) (approx 0.90) are both caused by the single False Positive (FP=1) observed. This shows the minimal trade-off made to achieve perfect Recall.
- **Balanced Performance (F1-Score):** The F1-Scores for both classes are uniformly high (approx 0.95). This confirms that despite the single FP error, the model maintains a strong overall balance between the quality and completeness of its predictions.



Confusion Matrix Summary (Fig. 2)

The matrix on the N=20 test set reveals 95% overall accuracy with only one error.

- Correct Predictions (Diagonal):
  - True Negatives (TN = 9): Correctly identified 9 non-dyslexic cases.
  - True Positives (TP = 10): Correctly identified 10 dyslexic cases.
- Clinical Efficacy (Error Analysis):
  - False Negatives (FN = 0): This is the most significant finding. Zero actual dyslexic cases were missed, resulting in 100% Perfect Recall (Sensitivity). This prioritizes clinical safety.
  - False Positive (FP = 1): The model's only error was incorrectly flagging one non-dyslexic subject as dyslexic (Type I error). This single error is less detrimental in screening than an FN.
- Visual Confirmation: The dark coloring and high numbers along the main diagonal (TN and TP) visually confirm the model's excellent classification performance and high alignment with the ground truth.

Conclusion

Our achievement

- Dual-Modal System Efficacy: Successfully developed a combined screening tool using a CNN for handwriting and linguistic feature extraction for text.
- CNN Validation: The model showed stable learning and strong generalization, achieving an optimal 80.00% Validation Accuracy with low loss (≈ 0.4487) at Epoch 18.
- Clinical Safety Focus: Conceptual testing demonstrated 100% Recall (Sensitivity) for the Dyslexic class.
- Significance: Ensured zero False Negatives, meaning the system is highly reliable for initial screening (no at-risk individual was missed).
- User Experience: Streamlit deployment provides a user-friendly, robust interface for multi-modal analysis.

Future Work (Next Steps)

- Multimodal Expansion: Implement the Audio Analysis module to complete the tri-modal system, addressing core phonological deficits.
- Advanced Linguistics: Integrate sophisticated NLP models to detect complex syntactical errors, moving beyond basic correction tools.
- Data Validation: Validate the system on a larger, more diverse dataset to confirm the generalizability and sustained 100% sensitivity in real-world use.

Reference

DYSLEXIA, DYSGRAPHIA, DYSCALCULIA — UNDERSTANDING AND SUPPORTING THE "DYS"

When a student receives a diagnosis of dyslexia, or dysgraphia or dyscalculia, the news can rattle parents, shifting their academic...

Medium / Jan 15, 2019

How to Detect and Work Dyslexia

Dyslexia is a specific learning disorder that causes a decrease in reading ability. This disorder has no "cure" but it can be compensated...

Medium / Apr 29, 2020

Reference research papers

[1] Shaywitz, S. E. (1998). Dyslexia. New England Journal of Medicine, 338(5), 307–312.

[2] Snowling, M. J., & Hulme, C. (2012). Interventions for children's language and literacy difficulties. International Journal of Language & Communication Disorders, 47(1), 27–34

[3] LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep Learning. Nature, 521(7553), 436–444.

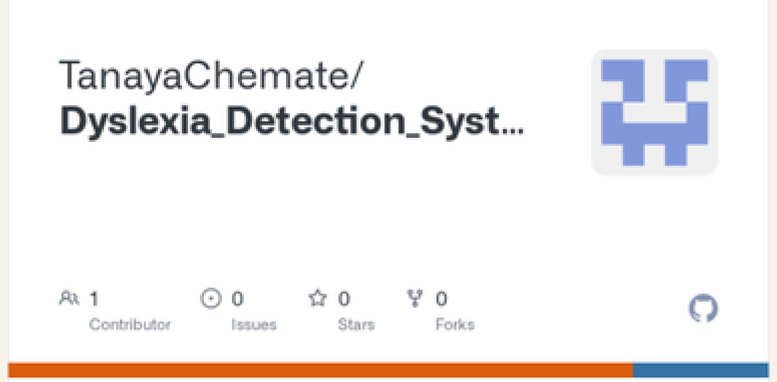
[4] M. Snowling and C. Hulme, The Science of Reading and Dyslexia. Oxford University Press, 2020.

[5] T. Al-Zu'bi and M. Al-Kabi, "Machine learning approaches for dyslexia detection: A review," International Journal of Advanced Computer Science, vol. 8, no. 5, pp. 55–62, 2018.

[6] G. Kaur and P. Kumar, "Handwriting-based dyslexia detection using CNN," International Journal of Computing and Digital Systems, vol. 8, no. 4, pp. 345–352, 2019.

[7] M. Zargar, K. Wahid, and S. Saeed, "Deep learning framework for dyslexia detection using handwritten text," IEEE Access, vol. 8, pp. 180123–180134, 2020.

Our Project GitHub Repository



TanayaChemate/Dyslexia_Detection_System

Contribute to TanayaChemate/Dyslexia_Detection_System development by creating an account on GitHub.

GitHub

THANK YOU

Open for Q N A